A Child with Loose Stool

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A child with loose stool – Who? Where?

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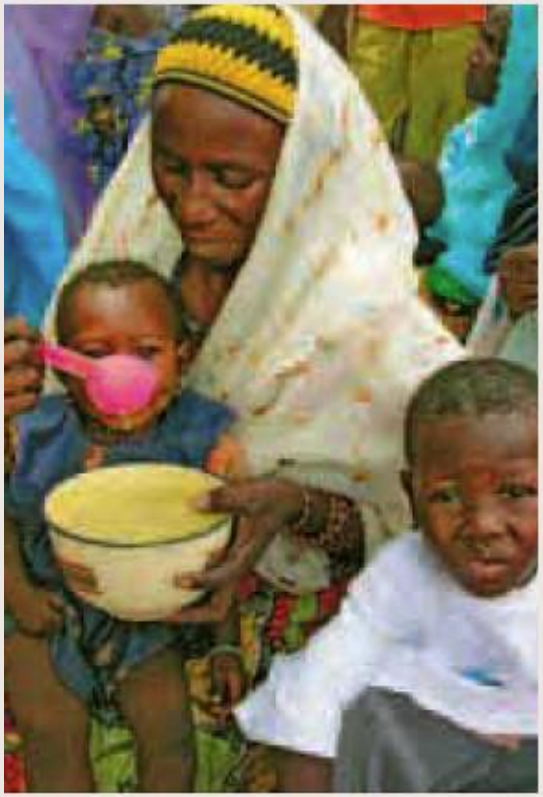

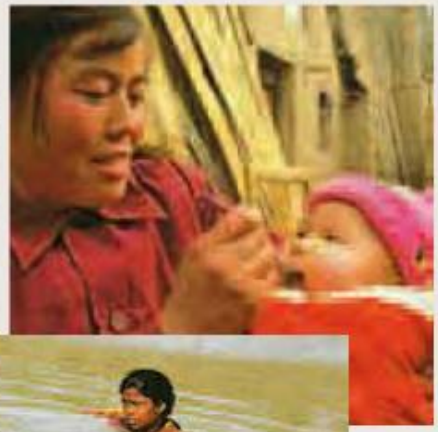

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Gastroenteritis in children

Etiology and management vary with geographical region, socioeconomic situations and health service provision

BUT

Majority of principles hold true

History Taking

1: Clarification of Chief Complaint

- Is this 'diarrhea'?

  Alteration in normal bowel movement characterized by change in stool consistency to loose or watery, with increase in stool frequency and volume

- Character: watery vs bloody, presence of mucus

  Watery stool without blood or mucus: usually caused by enterotoxin, virus or protozoan

  Bloody stool with mucus: cytotoxin-producing bacteria causing mucosal inflammation, enteric parasite

- Is this an acute or chronic problem?

Differential diagnosis of acute-onset bloody stool, fever and abdominal cramps:
- Campylobacter
- Shigella
- Salmonella
- Clostridium difficile
- Yersinia enterocolitis
- Vibrio parahemolyticus
- Enteroinvasive E. coli
- Enterohemorrhagic E. coli (O157:H7)
- Initial presentation of inflammatory bowel disease

2: Severity Assessment from the History

1. No. of episodes
2. Frequency
3. Estimated amount as judged by the parents

3: Any other concurrent symptoms

- Vomiting
- Fever
- Poor appetite: quantify the intake
- Abdominal pain
- ‘Irritability’

4: Is this child at risk of Dehydration?

Symptoms of dehydration

- Dry lips
- Cool limbs
- Urine output

Risk factors for dehydration

- Frequent profuse watery diarrhea
- Vomiting
- Poor fluid intake
- Concurrent fever

5: Formulation of Differential Diagnosis

Acute diarrhea

- Infective:

  - bacterial vs virus

  - secretory vs inflammatory
- Food intolerance
- Osmotic agents
- Drugs

Chronic diarrhea

6: ‘Why me?’

- Infective cause: what is the source?

  - usual intake
  - change in content
  - change in preparation: the person, the method, the utensils
  - ‘unhygienic food’ e.g. undercooked food, raw food, eating out
  - known contacts with persons having similar symptoms
  - nursery
  - Occupations of family members: cooks, kitchen workers, health care workers

Epidemiological and etiological considerations

- Setting: child-care centers, institutions, hospitals
- Family members, close contacts
- Food-borne or water-borne diarrhea occurring after common exposure
- Antimicrobial-associated diarrhea
- Diarrhea of travelers
- Diarrhea in immunocompromised hosts
- History of chronic diarrhea or underlying GI diseases

7: Clarify what has been done

- Did the mother give him sufficient fluid?
- How well does he tolerate?
- Did the mother give him any medications?
- Was the child seen by a doctor already? What has been done?

8: Past Health, and other related history

- Past health: recurrent diarrhea?
- Growth
- Diet pattern
- Birth history
- Immunization
- Family history
- Social history

Acute infective gastroenteritis - Etiology

- Rotavirus is the most common cause of GE in children, across community and hospital-based studies
- The most frequent pathogens causing bacterial GE are campylobacter, salmonella and E. coli
- The most common protozoal GE is caused by cryptosporidium
- Enteric pathogens commonly associated with traveling:
  - Most common: Salmonella
  - Others: campylobacter, E. coli, cryptosporidium
  - Shigella, giardia, entamoeba (Indian subcontinent, sub-Saharan, southern Africa)

Acute infective gastroenteritis - Etiology

Virus	Rotavirus Norovirus Adenovirus group F	Calicivirus Astrovirus Small round structured virus
Bacteria	E. coli - O157 - Non-O157 Salmonella spp. Shigella spp. Campylobacter spp.	Clostridium difficile Clostridium perfringens Staphylococcus aureus Bacillius spp. Vibrio spp. Yersinia spp. Aeromonas spp.
Protozoa	Cryptosporidium parvum Giardia lamblia	

Acute gastroenteritis – Duration and Natural History

- Diarrhea usually lasts for 5-7 days, in most stops within 2 weeks
- Vomiting usually lasts for 1-2 days
- Factors associated with more prolonged course of GE:

  - Co-infection of enteric pathogens

  - Bloody and / or mucoid stools

  - Malnutrition

  - Indiscriminate use of antibiotics

Clinical Assessment

A child with diarrhea and / or vomiting –
What characteristics may suggest a diagnosis other than GE?

Table 3.3 Key symptoms and signs in the differential diagnosis of children presenting with diarrhoea and/or vomiting[a]

Alternative diagnosis	Key symptoms	Key signs[b]
Non- enteric infections:		
• pneumonia	Cough, shortness of breath, chest pain	Tachypnoea, tachycardia
• urinary tract infection	Frequency and dysuria	
• meningitis	Persistent vomiting, altered consciousness, irritability, photophobia	Petechial purpuric rash, neck stiffness, bulging fontanelle in infants
• acute otitis media	Earache	
• toxic shock syndrome	Non-specific muscle aches, faintness	Clinical shock, red non-specific rash, possible site of bacterial entry such as small burn or injury
Non-infective gastrointestinal disorders:		
• ulcerative colitis, Crohn's disease or coeliac disease	Prolonged diarrhoea (>14 days), bloody diarrhoea	Failure to thrive or weight loss
Surgical disorders:		
• Bowel obstruction, intussusceptions or ischaemic bowel	Bilious vomiting, severe or localised abdominal pain, bloody diarrhoea	Abdominal distension, rebound tenderness, mucoid/bloody stools
Drug-related:		
Review drug history (e.g. antibiotic therapy)		

[a] Children younger than 12–18 months commonly present with non-specific symptoms and signs of non-enteric infections and non-infective gastrointestinal disorders.

[b] High fever (temperature ≥ 38 °C for children younger than 3 months and ≥ 39 °C for children 3 months or older) may occur in gastroenteritis, but may also be a pointer to non-enteric infections.

NICE, 2009

Assessing dehydration and shock

Recommendations on risk factors for dehydration

Recognise that the following are at increased risk of dehydration:

- children younger than 1 year, particularly those younger than 6 months
- infants who were of low birthweight
- children who have passed more than five diarrhoeal stools in the previous 24 hours
- children who have vomited more than twice in the previous 24 hours
- children who have not been offered or have not been able to tolerate supplementary fluids before presentation
- infants who have stopped breastfeeding during the illness
- children with signs of malnutrition.

NICE, 2009

Assessing dehydration and shock

Table 4.5 Classification of dehydration severity by the American Subcommittee on Acute Gastroenteritis[5]

Variable	Mild, 3–5%	Moderate, 6–9%	Severe, ≥10%
Blood pressure	Normal	Normal	Normal to reduced
Quality of pulses	Normal	Normal or slightly decreased	Moderately decreased
Heart rate	Normal	Increased	Increased
Skin turgor	Normal	Decreased	Decreased
Fontanelle	Normal	Sunken	Sunken
Mucous membrane	Slightly dry	Dry	Dry
Eyes	Normal	Sunken orbits	Deeply sunken orbits
Extremities	Warm, normal capillary refill	Delayed capillary refill	Cool, mottled
Mental status	Normal	Normal to listless	Normal to lethargic or comatose
Urine output	Slightly decreased	<1 ml/kg per hour	<< 1 ml/kg per hour
Thirst	Slightly increased	Moderately increased	Very thirsty or too lethargic to indicate

	Increasing severity of dehydration →		
	No clinically detectable dehydration	Clinical dehydration	Clinical shock
Symptoms (remote and face-to-face assessments)	Appears well	Appears to be unwell or deteriorating	–
	Alert and responsive	Altered responsiveness (for example, irritable, lethargic)	Decreased level of consciousness
	Normal urine output	Decreased urine output	–
	Skin colour unchanged	Skin colour unchanged	Pale or mottled skin
	Warm extremities	Warm extremities	Cold extremities
Signs (face-to-face assessments)	Alert and responsive	Altered responsiveness (for example, irritable, lethargic)	Decreased level of consciousness
	Skin colour unchanged	Skin colour unchanged	Pale or mottled skin
	Warm extremities	Warm extremities	Cold extremities
	Eyes not sunken	Sunken eyes	–
	Moist mucous membranes (except after a drink)	Dry mucous membranes (except for 'mouth breather')	–
	Normal heart rate	Tachycardia	Tachycardia
	Normal breathing pattern	Tachypnoea	Tachypnoea
	Normal peripheral pulses	Normal peripheral pulses	Weak peripheral pulses
	Normal capillary refill time	Normal capillary refill time	Prolonged capillary refill time
	Normal skin turgor	Reduced skin turgor	–
	Normal blood pressure	Normal blood pressure	Hypotension (decompensated shock)

Within the category of 'clinical dehydration' there is a spectrum of severity indicated by increasingly numerous and more pronounced symptoms and signs. For clinical shock, one or more of the symptoms and/or signs listed would be expected to be present. (–) indicate that these clinical features do not specifically indicate shock. Symptoms and signs with red flags may help to identify children at increased risk of progression to shock. If in doubt, manage as if there are symptoms and/or signs with red flags.

Hypernatremic dehydration

- Dehydration with plasma Na > 150mmol/L
- ‘Doughy skin’ not reliable to identify patients with hypernatremia dehydration
- Severity of dehydration more often underestimated compared with normo-natremic dehydration
- Suspect hypernatremia dehydration if there are any of the following:

  - Jittery movement
  - Increased muscle tone
  - Hyepr-reflexia
  - Convulsions
  - Drowsiness or coma

“Hmmm….does this child need admission?”

Practical considerations

When should a child with gastroenteritis be hospitalized?

- Red flag 🚩 signs for significant dehydration
- Risk factors for significant ongoing loss, especially with severe vomiting compromising the chance of success with oral rehydration
- Patients with grossly bloody stool
- Patients with immunocompromised states

A child with vomiting and diarrhea….. GE……Should I do blood taking?

Are laboratory investigations necessary to assess dehydration?

- Do not routinely perform blood biochemical testing
- Measure plasma Na, K, urea, creatinine & glucose if:
  - IV fluid therapy is required
  - there are symptoms / signs that suggest hypernatremia
  - infant with unexplained drowsiness: rule out hypoglycemia
- Measure venous blood acid-base status and chloride concentration if shock is suspected or confirmed

Acidosis in GE

Causes:

- Loss of bicarbonate in diarrhea
- Lactic acidosis in hypovolemic shock

To distinguish: measure anion gap

- Raised anion gap: impaired tissue perfusion
- Normal anion gap: bicarbonate loss

Stool microbiological investigations – Are they necessary?

- Most acute GE in children are viral in origin
- Even if the illness is caused by bacterial enteric infection, most children do not require anti-microbial treatment
- Identification of a specific pathogen is not generally required

Stool microbiological investigations – Are they necessary?

Stool microbiological investigation is particularly useful in the following circumstances:

- In a child with diarrhea and systemic illness: as part of ‘sepsis workup’ before administration of empiric antibiotic therapy
- Presence of blood and mucus in stool
  - ♦ Bacterial / amoebic dysentery would require anti-microbial therapy
  - ♦ Enterohaemorrhagic *E. coli* O157:H7 is associated with hemolytic uremic syndrome
  - ♦ Antibiotic-related pseudomembranous colitis, caused by Clostridium difficile
  - ♦ Non-infective conditions e.g. inflammatory bowel disease should be considered if stool culture repeatedly negative
- The returning traveler: spectrum of pathogens differ
- Immunocompromised hosts
- Diarrhea with prolonged course

“Doctor, does my child need a drip?”

Should we give oral rehydration or intravenous fluid?

Fluid management

- Primary prevention of dehydration in children with GE but without clinical dehydration
- Rehydration in children with GE with dehydration
  - Oral rehydration therapy
  - IV fluid therapy

Fluid management

In children with GE but without clinical dehydration:

- Continue breastfeeding and other milk feeds
- Encourage fluid intake
- Discourage consumption of fruit juices and carbonated drinks, especially in those at increased risk of dehydration
- Offer oral rehydration solution (ORS) as supplemental fluid to those at increased risk of dehydration

Fluid management

In children with gastroenteritis and dehydration:

- ORS as the first choice for rehydration
- Use ORS for replacing fluid deficit in the first 3-4 hour of rehydration phase
- Continue breastfeeding and milk feeds
- Avoid fruit juices or carbonated drinks

ORS solutions

	WHO ORS solution pre-2002	WHO ORS solution post-2002	QMH rice-based ORS	ORS (commercial preparation)
Osmolarity (mOsm/L)*	311	245	n/a	
Solute (mmol/L)				
Sodium	90	75	50	
Potassium	20	20	20	
Chloride	80	65	40	
Bicarbonate	30	-	-	
Citrate	-	10	10	
Glucose	111	75	50**	

* Low osmolarity is considered as < 250mOsm/L

** Grams of rice powder / liter solution

- Compared with high-osmolarity ORS, the use of low-osmolarity ORS solution reduces the need for unscheduled IV fluid for treatment of dehydration in children with diarrhea
- Low-osmolarity ORS also leads to greater reduction in stool output and vomiting

Rice-based ORS

- At the brush border of the intestinal villi, rice starch is broken down into glucose by polysaccharidase and disaccharidase
- Na-glucose co-transporter promotes absorption of salts and water into the lumen of intestinal villi

Evidence:

- Rice-based ORS is associated with reduction in duration of diarrhea in children, and the effect is more significant in those with cholera

Oral rehydration strategy

- Principle: small, frequent feeds
- To replace fluid deficit over 4 hours
- Ensure appropriate amount of maintenance fluid is administered to the child
- Monitor for ongoing loss and replace accordingly
- Assess for fluid tolerance
- May need to adjust strategy of rehydration according to the child's response to oral rehydration therapy
- ⚑ Reassess! Reassess! Reassess!

Example

Table 5.2 Worked example of an oral rehydration strategy in a 12-month-old child weighing 10 kg

Strategy	Volume	Rate
Fluid deficit for replacement over 4 hours	500 ml	125 ml/hour
Maintenance fluids over 24 hours[a]	1000 ml	40 ml/hour
Total volume for first 4 hours	660 ml	165 ml/hour
Aliquot volume if administered at 10 minute intervals		27.5 ml/10 minutes

[a] Maintenance fluids[95]

Table 5.3 Maintenance fluid volume requirements based on body weight

Weight (kg)	Volume per day	Volume per hour
0–10	100 ml/kg	4 ml/kg per hour
10–20	1000 ml + 50 ml/kg for each kg > 10	40 ml + 2 ml/(kg > 10)
20+	1500 ml + 20 ml/kg for each kg > 20	60 ml + 1 ml/(kg > 20)

NICE, 2009

Intravenous fluid

Use IV fluid therapy for rehydration if:

- Signs of shock
- Impaired conscious state
- A child with red flag symptoms or signs, showing clinical evidence of deterioration despite oral rehydration therapy
- Failure of oral rehydration e.g. intolerance due to persistent vomiting
- Paralytic ileus

Treatment of Shock

Shock

↓

Rapid IV infusion of 0.9% NaCl, 20ml/kg

↓

Persistent signs of shock

↓

Another Rapid IV infusion of 0.9% NaCl, 20ml/kg

Consider possible causes of shock other than dehydration

Signs of shock resolved ↙ ↘ Persistent signs of shock

Continue rehydration with IV fluid

Consult intensive care specialist

IV fluid rehydration

- WHO recommends IV rehydration to correct fluid deficit should be completed in 3-6 hours
- Attempt early and gradual introduction of oral rehydration therapy during IV fluid therapy
- If tolerated, stop IV fluid and complete rehydration with oral rehydration therapy

Treatment of hypernatremic dehydration

- Use isotonic solution e.g. 0.9% NaCl for fluid deficit replacement and maintenance
- Replace fluid deficit slowly – over 48 hours
- Monitor plasma sodium frequently, aiming to reduce it at a rate of less than 0.5mmol/L per hour

Fluid management following rehydration

- Encourage breastfeeding and other milk feeds
- Encourage fluid intake
- Prevent recurrence of dehydration by replacing ongoing loss from diarrhea with *additional* ORS e.g. 5-10ml/kg per large watery stool, for the following at-risk groups:

  - children < 1 year, especially those < 6 months
  - Infants who were born low birth weight
  - diarrheal stools > 5 in the previous 24 hours
  - vomit > 2 in the previous 24 hours
- Restart oral rehydration therapy if dehydration recurs after rehydration

“Doctor, can my child take milk?

Is diluted milk better?

Should we switch to soy formula?”

Nutritional management

- Re-introduce usual feeding early, including solid food
- No evidence that diluted milk is of benefit; give full-strength milk as usual
- Available evidence shows no benefit from special milk formulas, such as lactose-free or soy-based formula

“Doctor, can you give some medicine to stop my child’s vomiting and diarrhea?”

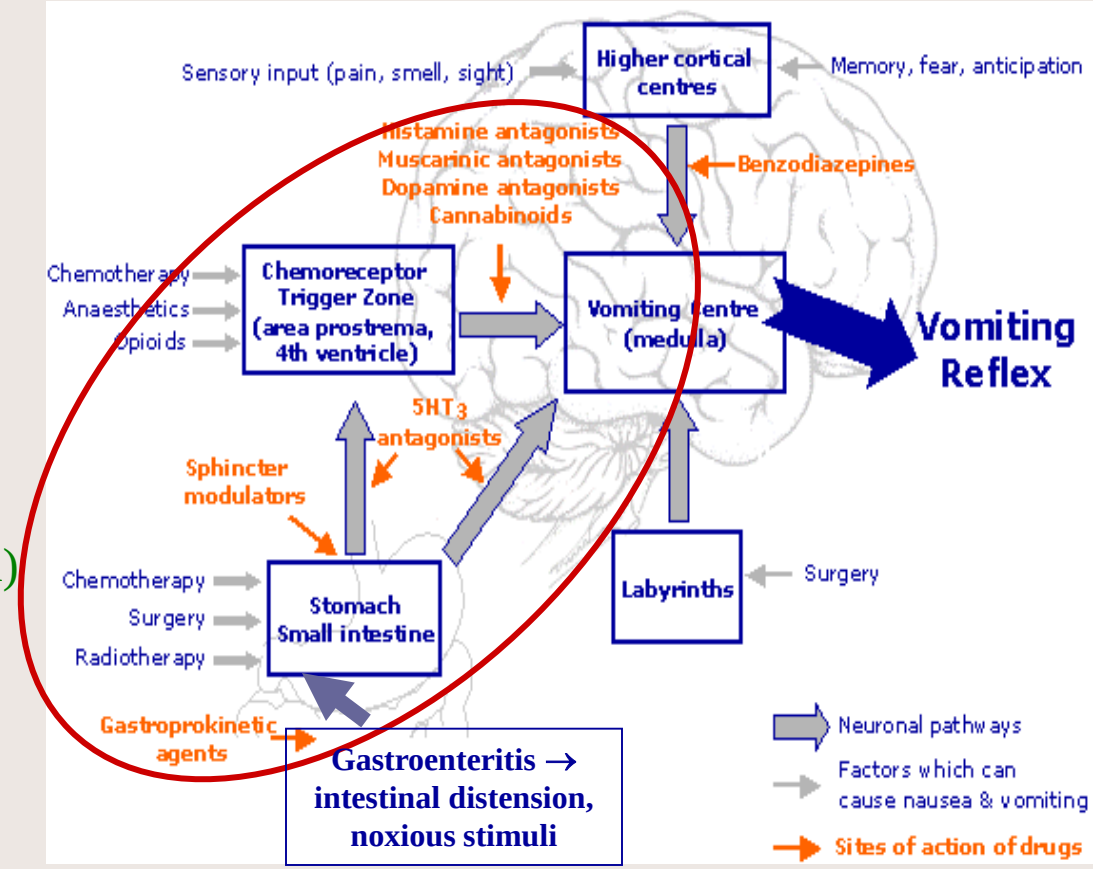
Anti-emetics

Anti-emetics commonly prescribed in medical practice:

Promethazine [phenergan]
anti-histamine H1, anti-cholinergic, anti-dopaminergic activity

Metoclopramide (maxolon)
Anti-dopamine D2, weak anti-5HT3 activity

Ondansetron (zofran)
Potent, highly-selective 5-HT3 receptor antagonists



Anti-emetics in managing vomiting in children with GE: what is the evidence?

- Oral ondansetron can reduce the occurrence of persistent vomiting, success of oral rehydration and thus decrease in the use of IV fluid from RCT trials
- However, the frequency of stool passage increases with ondansetron
- Postulated to be a result of retention of fluids and toxins that would otherwise have been eliminated through the process of vomiting
- Most children have self-limiting vomiting; routine use of ondansetron not recommended
- Use of other anti-emetics e.g. promethazine, metoclopramide have significant adverse effects e.g. dystonic reactions of facial and skeletal muscles, oculogyric crisis, do not use in children

Anti-diarrheal agents

- Commonly prescribed agents in medical practice:
  Adsorbents e.g. kaolin, smectite
  Anti-motility agents e.g. loperamide
- No high-level evidence available for the benefit of anti-diarrheals, not recommended
- Potentially serious adverse side-effects: abdominal distension, ileus, drowsiness in loperamide; increase in intestinal transit time reduce clearance of pathogens and toxins

Probiotics

‘‘A live microbial food ingredient that, when ingested in sufficient quantities, exerts health benefits on the consumer”

Possible mechanism:

- Compete with pathogens for binding sites and substrates
- ↓ intestinal luminal pH
- Upregulation of genes mediating immunity
- Production of trophic short-chin fatty acids to promote mucosal cell growth and differentiation
- Examples: lactobacillus, bifidobacillus, saccharomyces yeast
- Some evidence of possible clinical benefits (with specific formulations e.g. *Lactobacillus caseii GG)*, as a useful adjunct to rehydration therapy

“Doctor, will you treat my child with antibiotics?”

Antibiotics

- Most acute GE in children are self-limiting illness caused by viral infection, and does not require antibiotic treatment
- Give antibiotics when there is:

  - suspected or confirmed septicemia
  - extra-intestinal spread of bacterial infection
  - patient is < 6 months with salmonella gastroenteritis
  - patient is malnourished or immunocompromised with salmonella GE
  - *Clostridium difficile* associated pseudomembranous enterocolitis
  - giardiasis
  - dysentery caused by Shigella
  - amoebic dysentery
  - cholera

“Doctor, when can my child be discharged home?

Discharge plan

Time for discharge:

- Rehydration completed
- Minimal risk for recurrence of dehydration
- Uncomplicated disease course

Other considerations for admission / discharge:

- Reliable carer: capability of child care
- Understanding of the carer to your treatment plan: any previous experience in managing a child with GE, myths and misunderstandings

“Doctor, how should I take care of my child?”

Information and advice for parents and carers

- Most children with GE can be safely managed at home
- Empower parents and carers

Advice:

- Continue usual feeds, including breastfeeding or other milk feeds
- Encourage the child to drink plenty of fluids
- Discourage consumption of fruit juices and carbonated drinks
- Offer ORS solution as supplemental fluid, especially in the presence of ongoing loss
- Fluids should be given frequently in small portions

Information and advice for parents and carers

Watch out for symptoms of dehydration:

- appear to get more unwell
- altered responsiveness, irritability, lethargy
- Decreased urine output
- Pale, mottled skin
- Cold extremities

Should seek medical attention if symptoms of dehydration develops

Also seek medical attention if the disease course is atypical:

- usual duration of diarrhea is 5-7 days, in most children stops within 2 weeks
- usual duration of vomiting is 1-2 days, in most stops within 3 days

“My child *had* diarrhea……

He died.”

“Please save *this* child…”

Gastroenteritis – The global perspective

2010:
Global under-five mortality = 7.6 million
Diarrheal disease: 0.761 million (9.9%)

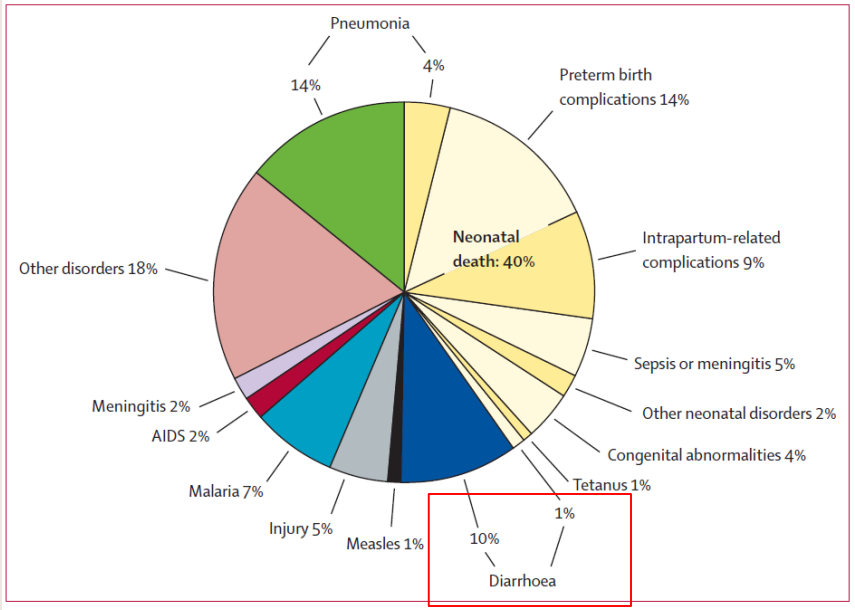


Figure 2: Global causes of childhood deaths in 2010

	Estimated number (UR; millions)
Neonates aged 0–27 days	
Preterm birth complications	1·078 (0·916–1·325)
Intrapartum-related complications	0·717 (0·610–0·876)
Sepsis or meningitis	0·393 (0·252–0·552)
Pneumonia*	0·325 (0·209–0·470)
Congenital abnormalities	0·270 (0·207–0·366)
Other neonatal disorders	0·181 (0·115–0·284)
Tetanus	0·058 (0·020–0·276)
Diarrhoea†	0·050 (0·017–0·151)
Children aged 1–59 months	
Other disorders	1·356 (1·112–1·581)
Pneumonia*	1·071 (0·977–1·176)
Diarrhoea†	0·751 (0·538–1·031)
Malaria	0·564 (0·432–0·709)
Injury	0·354 (0·274–0·429)
Meningitis	0·180 (0·136–0·237)
AIDS	0·159 (0·131–0·185)
Measles	0·114 (0·092–0·176)

Global, regional, and national causes of under-5 mortality in 2000–19: an updated systematic analysis with implications for the Sustainable Development Goals

Jamie Perin, Amy Mulick, Diana Yeung, Francisco Villavicencio, Gerard Lopez, Kathleen L Strong, David Prieto-Merino, Simon Cousens, Robert E Black, Li Liu**

Lancet Child Adolesc Health 2022; 6: 106–15

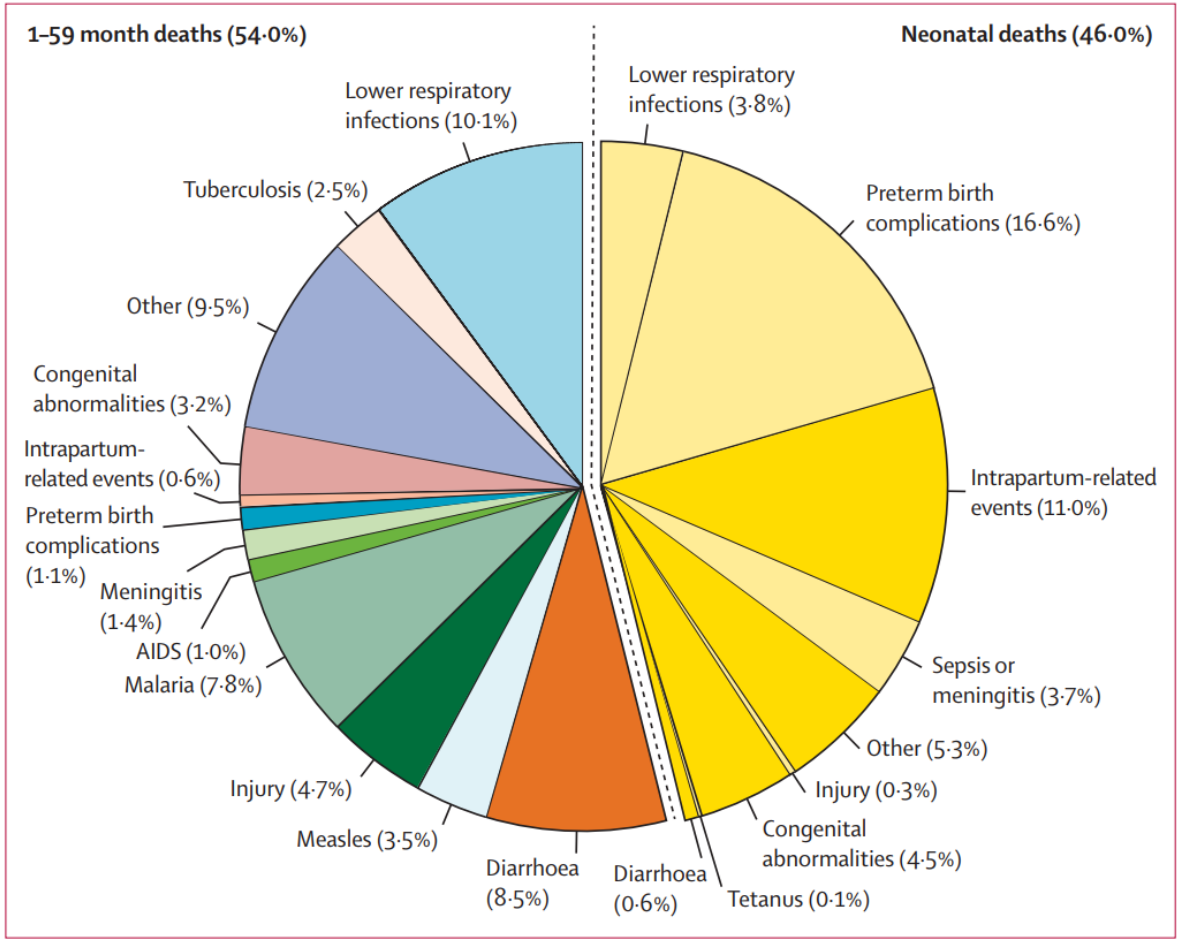


Figure 1: **Global causes of under-5 deaths in 2019**

In 2019, there were 5·30 million deaths among children younger than 5 years, primarily due to preterm birth complications (17·7%), lower respiratory infections (13·9%), intrapartum-related events (11·6%), and diarrhoea (9·1%), with 49·2% due to infectious causes. Vaccine-preventable deaths, such as for lower respiratory infections, meningitis, and measles, constituted 21·7% of under-5 deaths, and many other causes, such as diarrhoea, were preventable with low-cost interventions

Gastroenteritis – The global perspective

Gastroenteritis in the developed world

- Deaths associated with GE is rare
- Second most common cause of hospitalization and doctor visits
- Rotavirus accounts for 25-55% of all hospitalizations for GE

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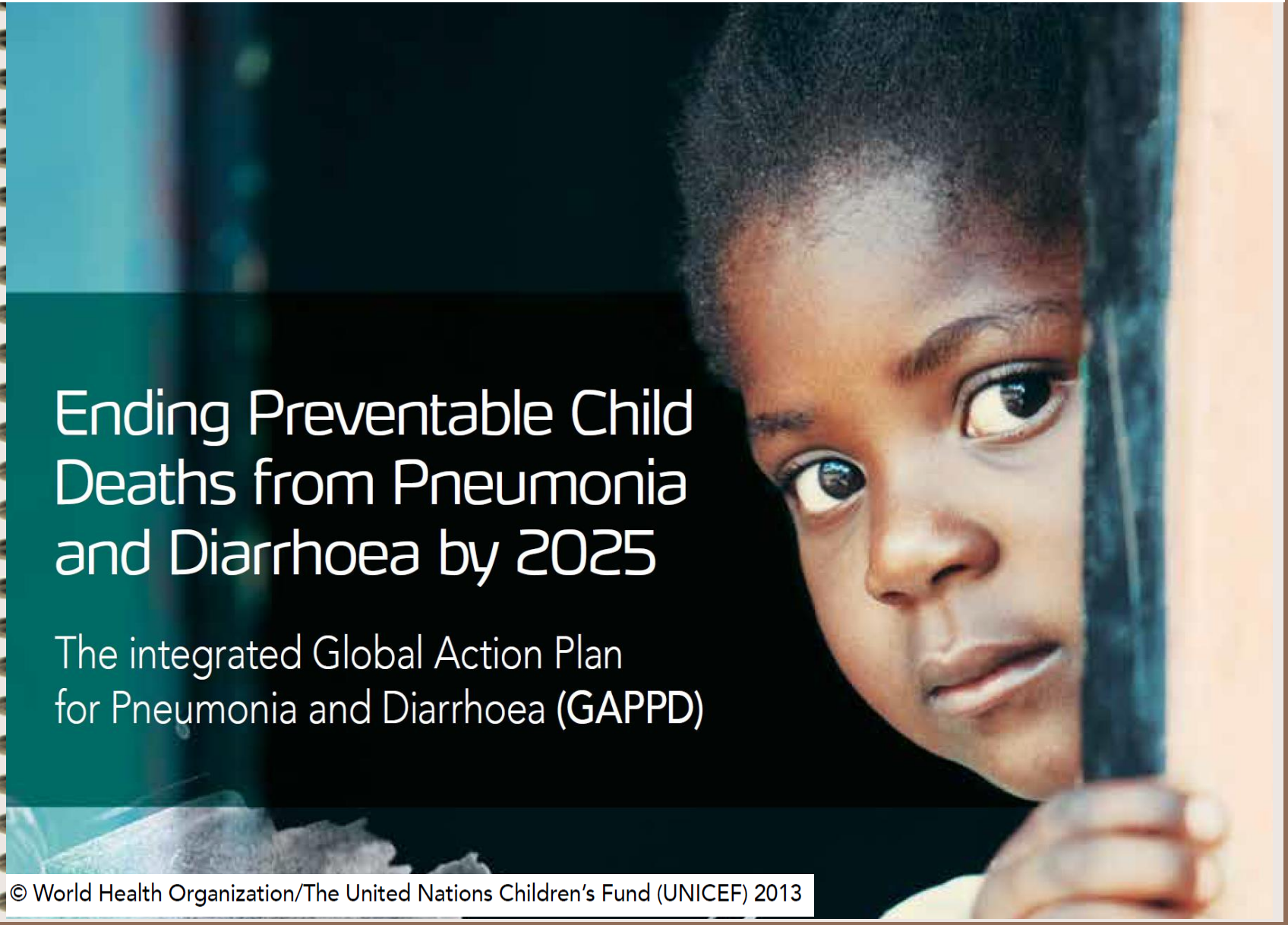
World Health Organization

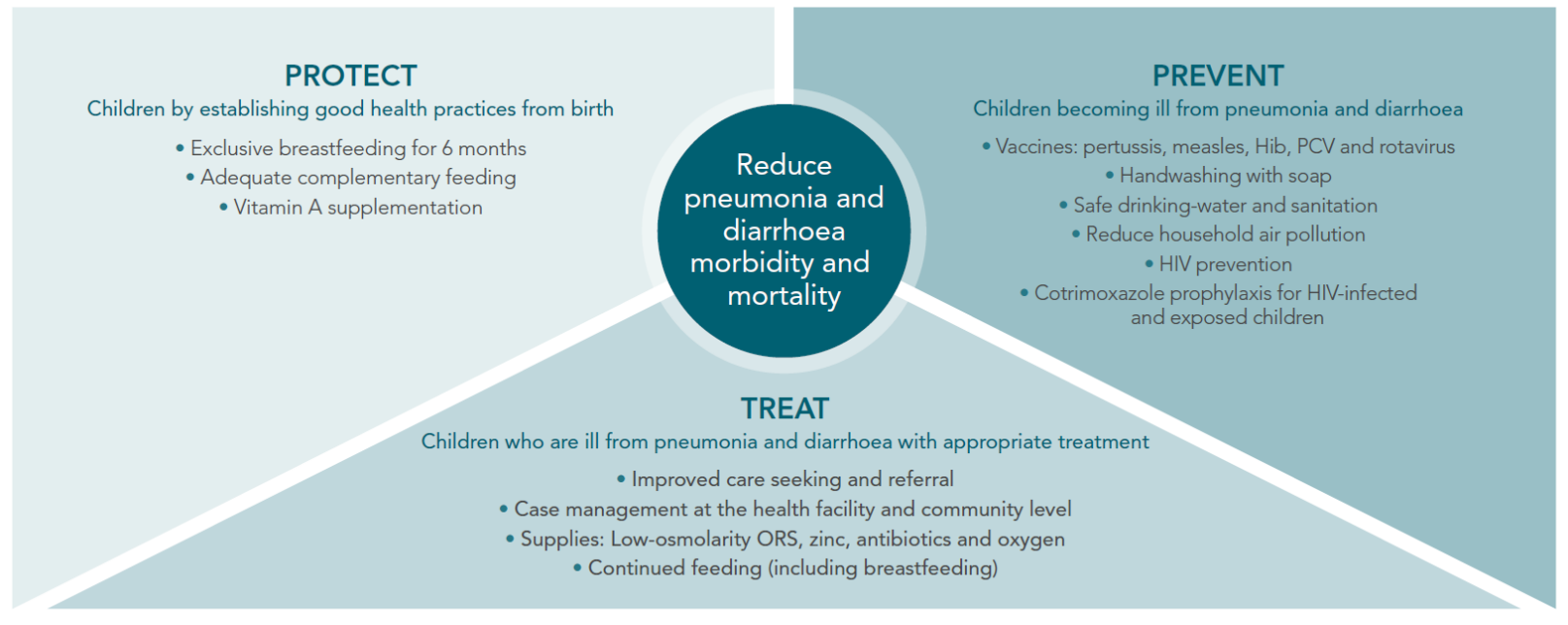


Ending Preventable Child Deaths from Pneumonia and Diarrhoea by 2025

The integrated Global Action Plan for Pneumonia and Diarrhoea (GAPPD)

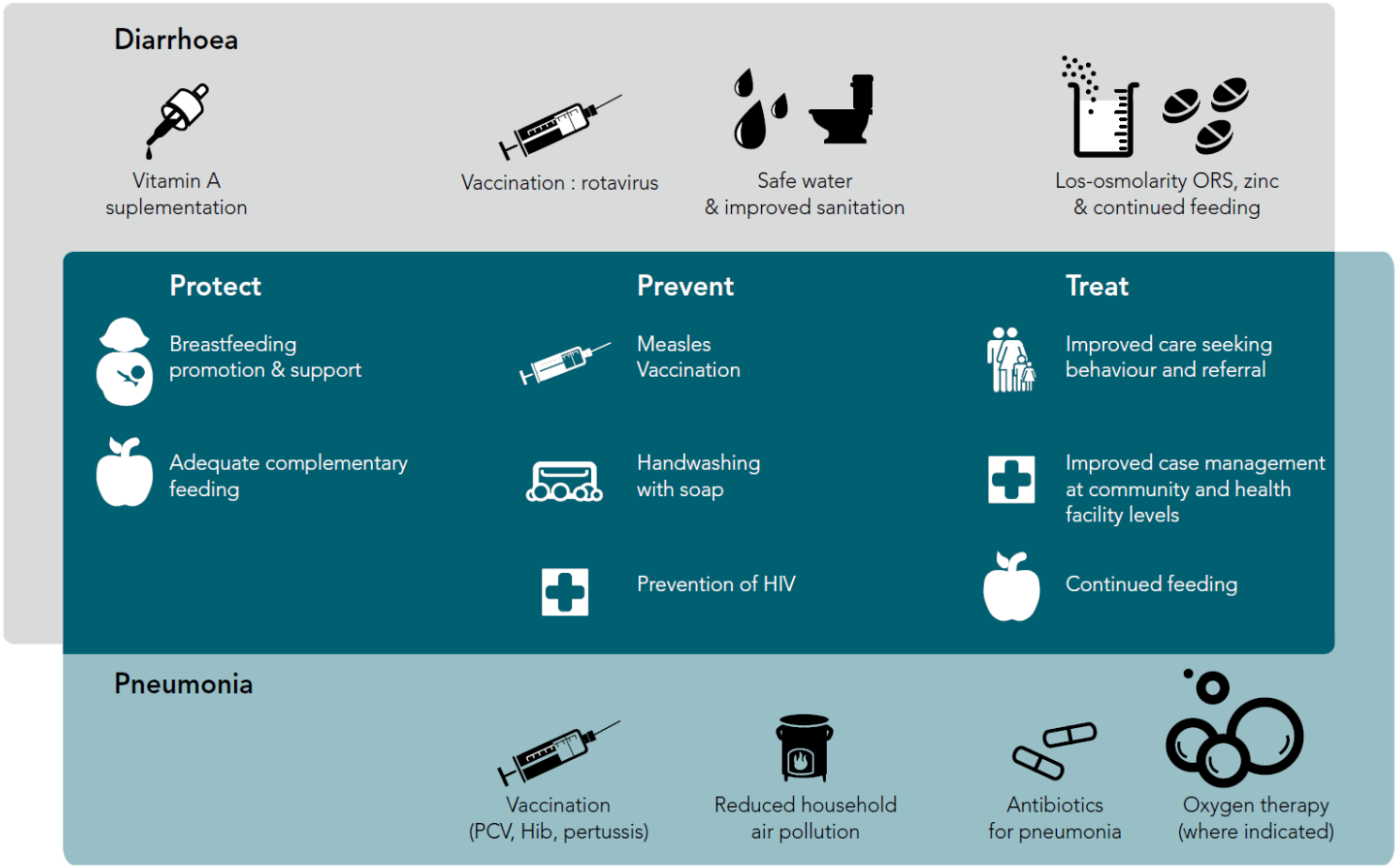
© World Health Organization/The United Nations Children's Fund (UNICEF) 2013

- Issues in developing countries:
  - Access to clean drinking water
  - Sanitation
  - Malnutrition



The integrated Global Action Plan for Pneumonia and Diarrhoea (GAPPD)

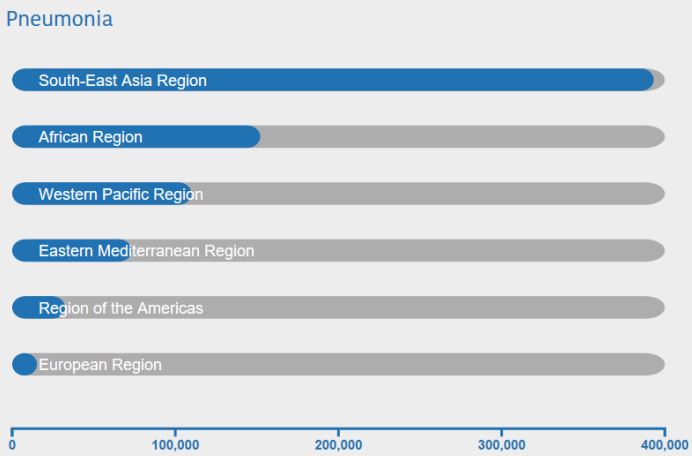
FIGURE 4. Complementarity of pneumonia and diarrhoea interventions

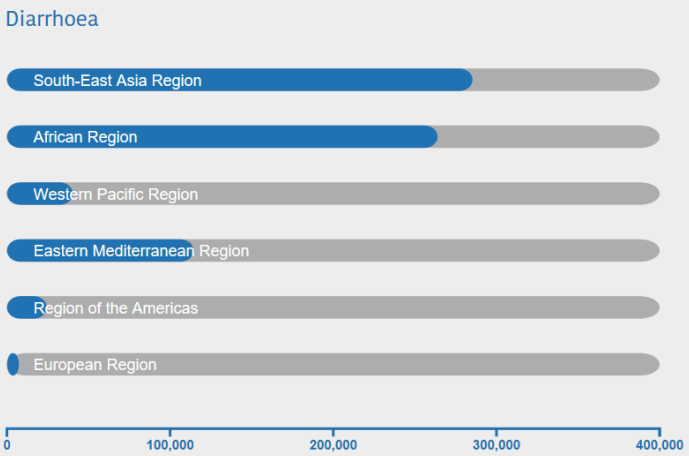


Source: Adapted PATH: Tackling the deadliest diseases for the world's poorest children.

The implementation of integrated approaches to protect, prevent and treat will contribute to ending preventable under-five deaths from pneumonia and diarrhoea by 2030

Decrease in under-five deaths due to pneumonia and diarrhoea in the period 2000-2015





Settings, personnel, needs, services, technology levels are different……

Principles are the same